DATA ENGINEERING ACADEMY | CAPSTONE PROJECT

Retail Sales Performance Analysis



AWS S3 / Snowflake / Snowpark Python / Streamlit

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Business question & dataset

Turning retail transaction records into measurable business performance

- **Goal**

  Identify leading products, regional differences, and monthly revenue patterns.

- **Source**

  sales.csv: 100 records, six columns, stored in AWS S3.

- **Fields**

  DATE, PRODUCT, REGION, UNITS_SOLD, UNIT_PRICE, SALES.

- **Scope**

  Descriptive analysis, not profit, causation, or demand forecasting.

End-to-end architecture

Separation of raw storage, analytical processing, and presentation



Snowflake reads the S3 CSV through an IAM-backed storage integration and stage.

Snowpark computes the metrics; Streamlit and Altair present the results.

Data validation & reliability

Checks were executed before the analytical aggregations

100

Rows inspected

0

Missing values

0

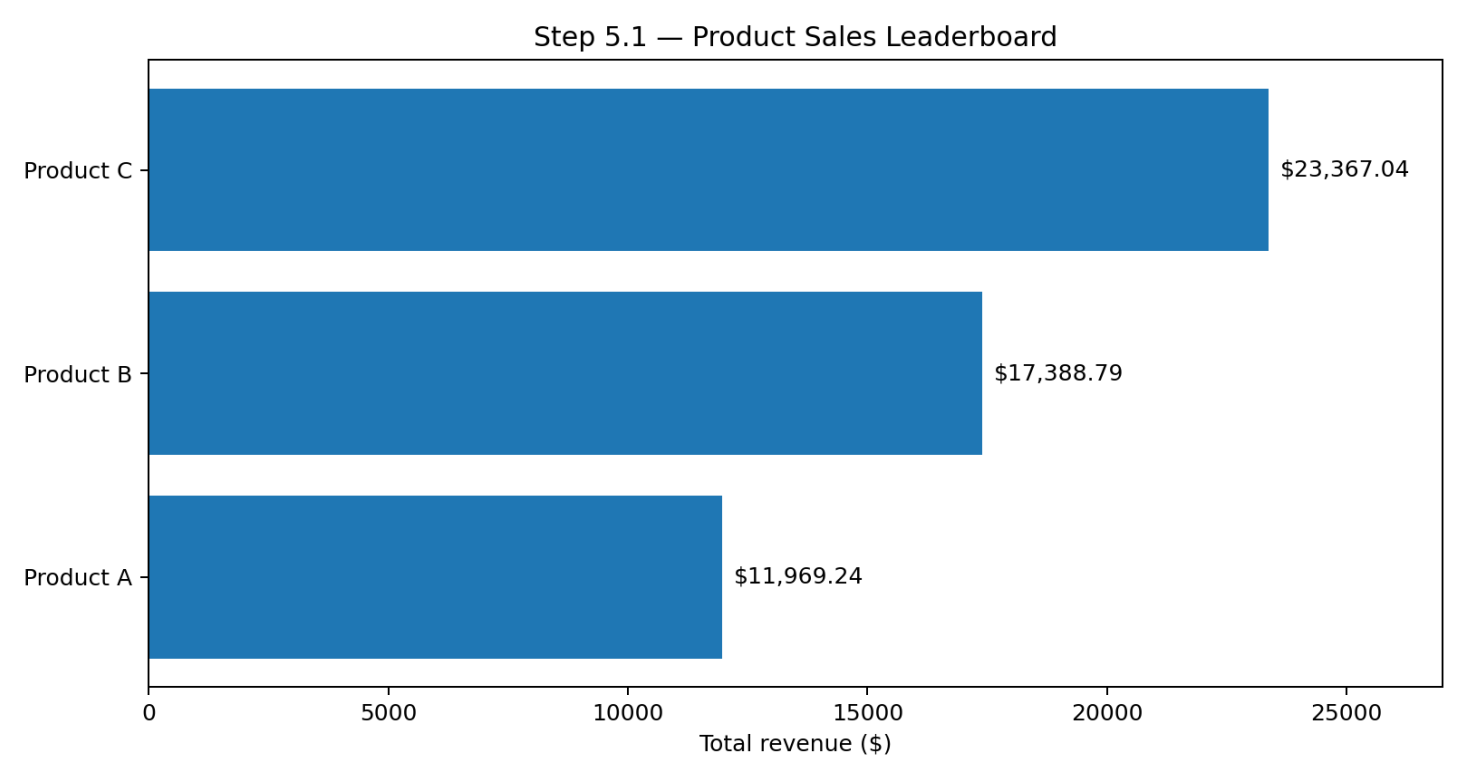
Exact duplicates

Verified checks support this dataset; they do not prove all possible source errors are absent.

Revenue calculations use UNITS_SOLD x UNIT_PRICE within Snowpark.

Product sales leaderboard

Product C leads in both revenue and units sold



$23,367.04

Product C | 425 units

$17,388.79

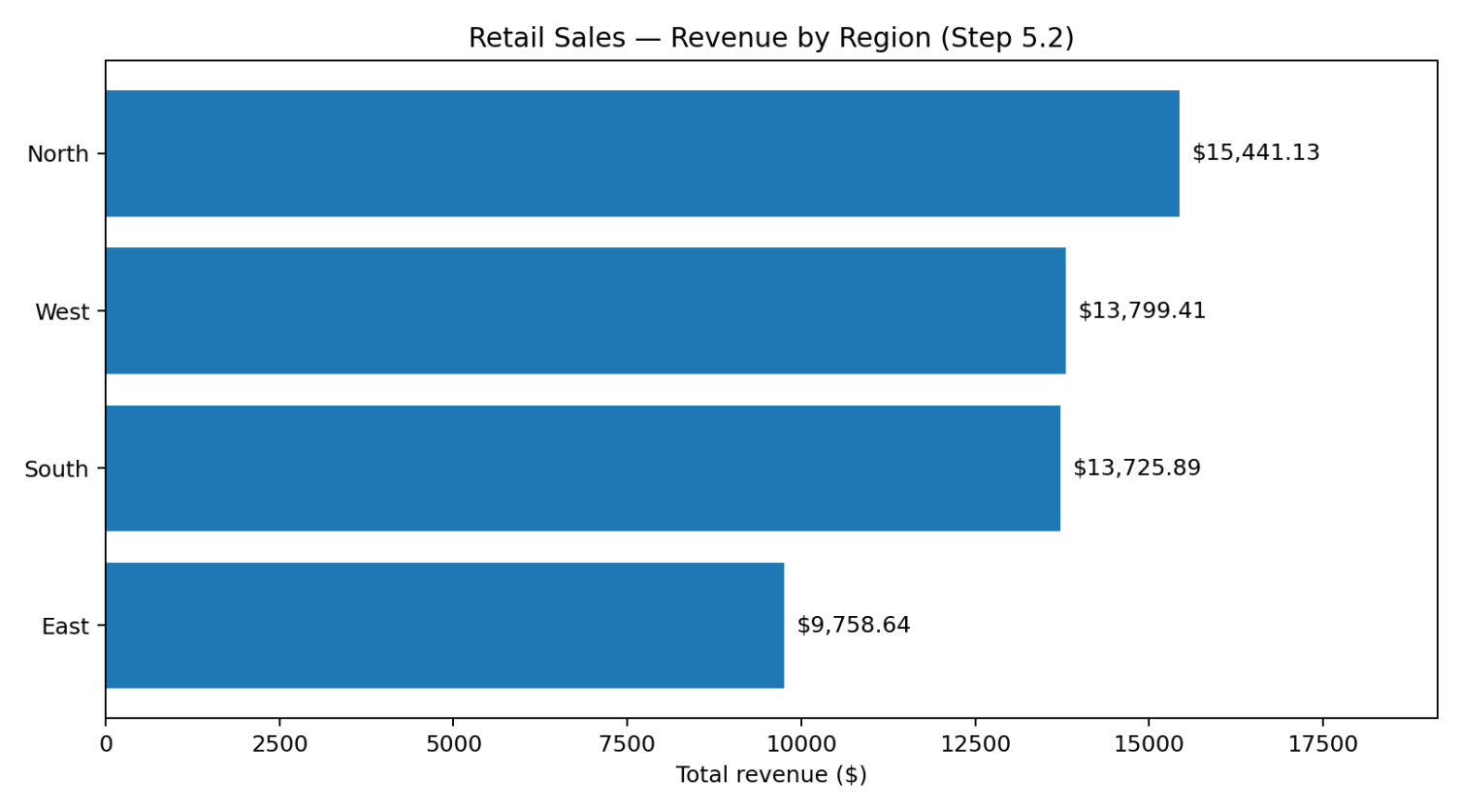
Product B | 333 units

$11,969.24

Product A | 252 units

Regional sales comparison

North generated the most revenue; East the least



North $15,441.13

West $13,799.41

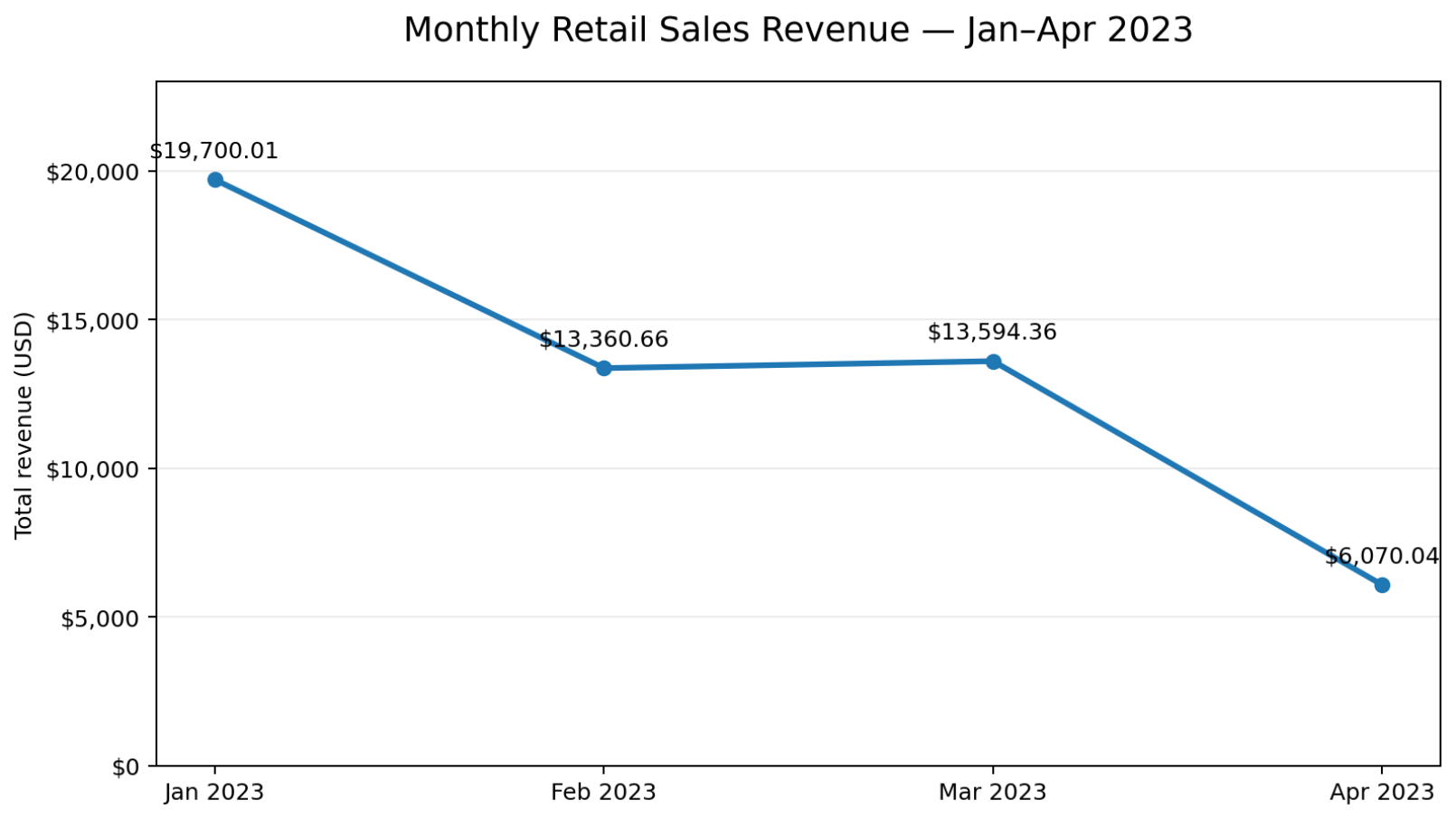
South $13,725.89

East $9,758.64

Regional revenue does not establish profitability.

Monthly revenue trend

April has partial coverage and should not be compared as a full month



Jan $19,700.01

Feb $13,360.66

Mar $13,594.36

Apr $6,070.04

April data through Apr 10.

Only four months available.

Units, revenue & average per unit

Five required metrics were calculated using Snowpark

$52,725.07

Total revenue

1,010

Total units

$52.20

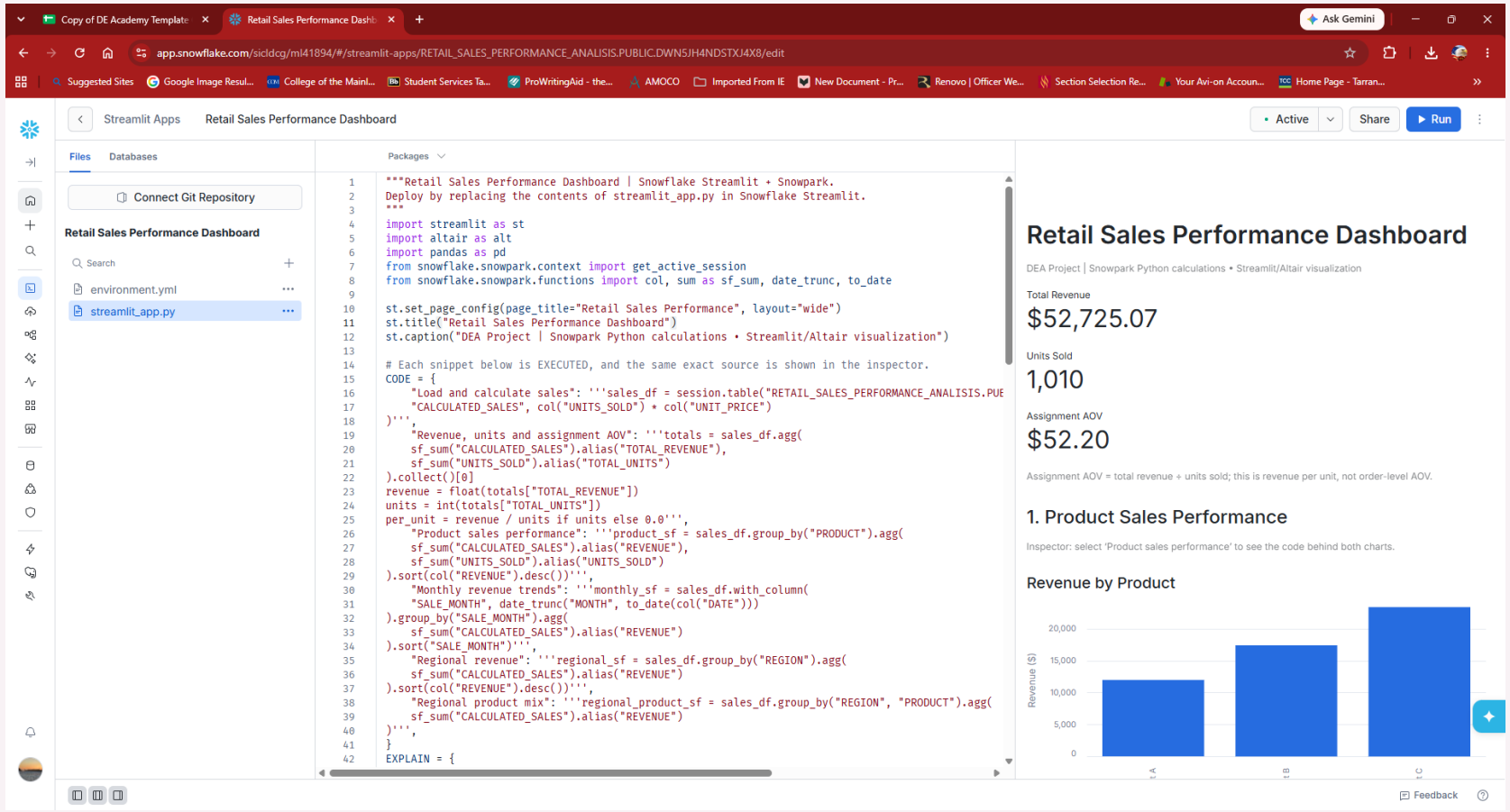
Revenue per unit

Assignment calls revenue / units sold "AOV"; this is technically average revenue per unit.

Conventional order-level AOV requires order identifiers that are not available.

Live Streamlit dashboard

Interactive KPI cards, sales charts, and reviewable Snowpark logic



Live Snowflake application (access may require Snowflake sign-in and permissions)

Snowpark Code Inspector

Designed for methodology transparency and evaluator review

- **What it shows**

  Selected KPI/chart -> matching Snowpark Python snippet and explanation.

- **Why it matters**

  Reviewers can trace calculations without searching through source files.

- **Separation of responsibilities**

  Snowpark calculates; Streamlit and Altair display the results.

Important: The inspector explains code; it is not independent proof of execution.

Why each technology was selected

A deliberate stack aligned with the assignment requirements

AWS S3	Preserves original CSV separately from analytics.
IAM + integration	Controlled S3 access without hardcoded AWS keys.
Snowflake	Central data storage, SQL checks, and computation.
Snowpark Python	In-database Python transformations and aggregations.
Streamlit + Altair	Interactive dashboard without a separate BI platform.
pandas	Display-ready conversion of small aggregated outputs.
GitHub	Versioned code, evidence, and project documentation.

Findings, limitations & delivery

A reproducible portfolio project with transparent methodology

KEY FINDINGS

Product C is the top seller; North is the highest-revenue region.

Total revenue: $52,725.07 across 1,010 units.

LIMITATIONS

No cost data or transaction IDs; April is incomplete.

DOCUMENTATION

README.md: overview and rationale

readme_analysis.md: metrics, charts, interpretation

readme_tech.md: setup, Snowpark code, validation, troubleshooting

AI assistance supported coding, debugging, and documentation; creator reviewed the work.